Molecular Microbiology (2001) **42**(3), 777–793

The LuxS-dependent autoinducer AI-2 controls the expression of an ABC transporter that functions in AI-2 uptake in *Salmonella typhimurium*

Michiko E. Taga, Julia L. Semmelhack and Bonnie L. Bassler*
Department of Molecular Biology, Princeton University, Princeton, NJ 08544-1014, USA.

Summary

In a process called quorum sensing, bacteria communicate with one another using secreted chemical signalling molecules termed autoinducers. A novel autoinducer called AI-2, originally discovered in the quorum-sensing bacterium *Vibrio harveyi*, is made by many species of Gram-negative and Gram-positive bacteria. In every case, production of AI-2 is dependent on the LuxS autoinducer synthase. The genes regulated by AI-2 in most of these *luxS*-containing species of bacteria are not known. Here, we describe the identification and characterization of AI-2-regulated genes in *Salmonella typhimurium*. We find that LuxS and AI-2 regulate the expression of a previously unidentified operon encoding an ATP binding cassette (ABC)-type transporter. We have named this operon the *lsr* (*luxS* regulated) operon. The Lsr transporter has homology to the ribose transporter of *Escherichia coli* and *S. typhimurium*. A gene encoding a DNA-binding protein that is located adjacent to the Lsr transporter structural operon is required to link AI-2 detection to operon expression. This gene, which we have named *lsrR*, encodes a protein that represses *lsr* operon expression in the absence of AI-2. Mutations in the *lsr* operon render *S. typhimurium* unable to eliminate AI-2 from the extracellular environment, suggesting that the role of the Lsr apparatus is to transport AI-2 into the cells. It is intriguing that an operon regulated by AI-2 encodes functions resembling the ribose transporter, given recent findings that AI-2 is derived from the ribosyl moiety of *S*-ribosylhomocysteine.

Accepted 9 August, 2001. *For correspondence. E-mail bbassler@molbio.princeton.edu; Tel. (+1) 609 258 2857; Fax (+1) 609 258 6175.

Introduction

Bacteria regulate gene expression in response to changes in cell population density by a process called quorum sensing. Specifically, quorum-sensing bacteria release and detect chemical signal molecules called autoinducers. Bacteria respond to the accumulation of a minimal threshold stimulatory concentration of autoinducer. Detection of autoinducers enables bacteria to distinguish between low and high cell population density and to control target gene expression in response to fluctuations in cell number. Quorum-sensing bacteria regulate processes that require the co-operation of a number of bacterial cells in order to be effective, and the individuals in the group profit from the activity of the entire assembly (Fuqua *et al*., 1996; Kleerebezem *et al*., 1997; Lazazzera and Grossman, 1998; Bassler, 1999; de Kievit and Iglewski, 2000; Miller and Bassler, 2001; Schauder and Bassler, 2001). These processes include bioluminescence, virulence, antibiotic production, sporulation and biofilm formation. Quorum sensing therefore allows a population of bacteria to co-ordinate behaviour and thus take on the characteristics of multicellular organisms.

In general, quorum sensing is controlled by acyl-homoserine lactone (HSL) autoinducers in Gram-negative bacteria and by modified oligopeptide autoinducers in Gram-positive bacteria (Lazazzera and Grossman, 1998; de Kievit and Iglewski, 2000; Miller and Bassler, 2001; Schauder *et al*., 2001). Gram-negative quorum-sensing circuits typically resemble the canonical circuit of *Vibrio fischeri*. Specifically, the acyl-HSL autoinducer synthase is similar to the *V. fischeri* LuxI enzyme, and a transcriptional activator similar to the *V. fischeri* LuxR protein is responsible for autoinducer recognition and target gene activation (Engebrecht *et al*., 1983; Engebrecht and Silverman, 1984; 1987; de Kievit and Iglewski, 2000; Miller and Bassler, 2001). In Gram-positive bacteria, the oligopeptide autoinducers are synthesized as precursor peptides that are processed, modified and subsequently secreted by ATP binding cassette (ABC)-type exporters. Most Gram-positive bacteria detect and respond to oligopeptide autoinducers via two-component phosphorylation cascades (Kleerebezem *et al*., 1997; Lazazzera and Grossman, 1998; Miller and Bassler, 2001). Both acyl-HSL and oligopeptide autoinducers are highly specific to the species that produce them, as autoinducers produced by one species usually do not influence the expression of genes in other species. It is remarkable that such exquisite signalling specificity exists in both types of quorum-sensing circuits, given the similarity in the members of each class of signal molecule.

Unlike all other quorum-sensing bacteria, *Vibrio harveyi*, a bioluminescent Gram-negative bacterium, uses a novel regulatory circuit to control quorum sensing. Specifically, *V. harveyi* controls density-dependent expression of the luciferase genes using a hybrid quorum-sensing circuit, with components common to both Gram-negative and Gram-positive quorum-sensing systems (Bassler, 1999; Freeman and Bassler, 1999a,b; Freeman *et al.*, 2000; Lilley and Bassler, 2000). *V. harveyi* uses an acyl-HSL autoinducer (called AI-1) as a signal molecule (Cao and Meighen, 1989; Bassler *et al.*, 1993). In addition, a second, novel autoinducer, termed AI-2, also regulates quorum sensing in *V. harveyi* (Bassler *et al.*, 1993; 1994a,b; 1997). It is hypothesized that *V. harveyi* uses AI-1 for intraspecies cell–cell communication and AI-2 for interspecies cell–cell signalling (Bassler *et al.*, 1997; Bassler, 1999; Surette *et al.*, 1999). These distinct signals presumably allow *V. harveyi*, which inhabits multispecies consortia, to vary its gene expression not only in response to changes in total cell number, but also in response to fluctuations in the species composition of the community.

Consistent with a role for AI-2 as a universal signal used for bacterial interspecies communication, over 30 species of Gram-negative and Gram-positive bacteria have now been shown to produce AI-2 (Bassler *et al.*, 1997; Surette and Bassler, 1998; Miller and Bassler, 2001). In every case, an AI-2 synthase that is highly homologous to the *V. harveyi* AI-2 synthase called LuxS is required for AI-2 production (Surette *et al.*, 1999). Recently, the biosynthetic pathway for AI-2 synthesis was described (Schauder and Bassler, 2001; Schauder *et al.*, 2001). AI-2 is produced from *S*-ribosylhomocysteine (SRH), a product

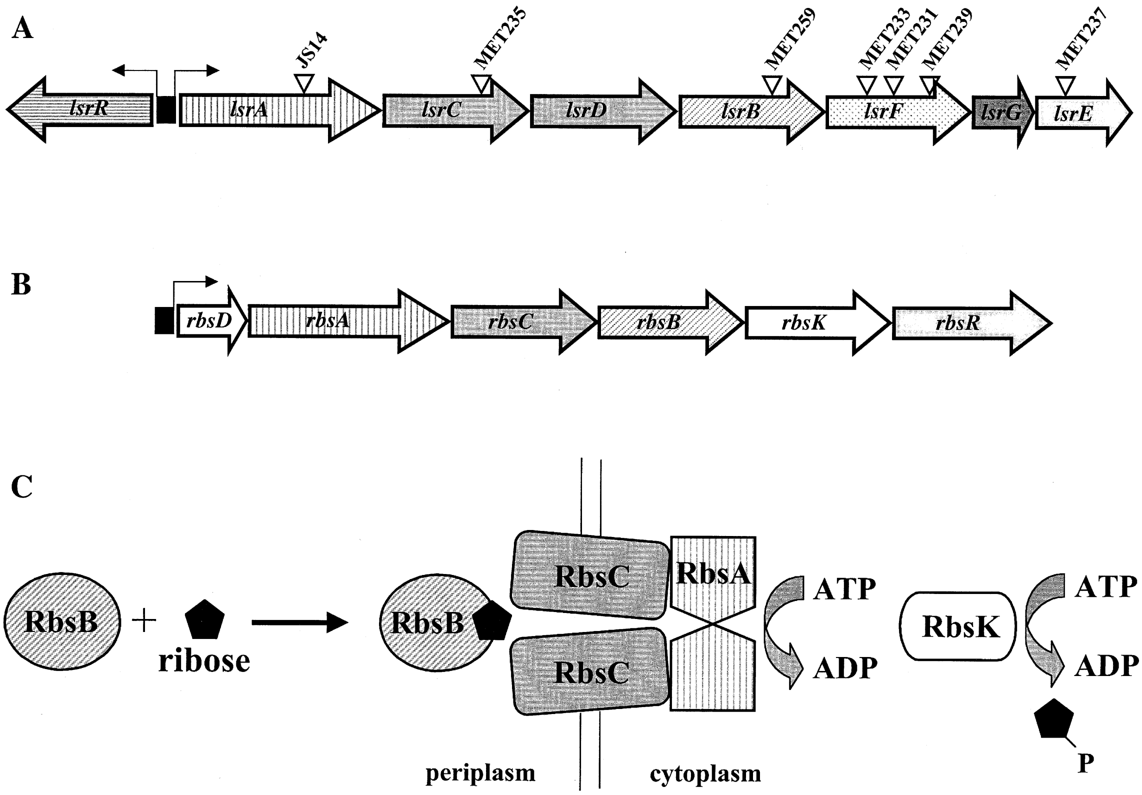

Fig. 1. The *lsr* and rbs operons of *S. typhimurium*.
A. The *lsr* operon of *S. typhimurium* is shown. This operon contains genes homologous to the transport component of the *rbs* operon (B). Triangles denote the sites of the MudJ insertions, with strain names above. A gene called *lsrR* is transcribed divergently from the *lsr* operon and encodes a protein (LsrR) required for AI-2 regulation of the *lsr* operon (see text).
B. The *rbsDACBKR* operon of *E. coli* and *S. typhimurium* encodes proteins required for the high-affinity transport and phosphorylation of ribose. *rbsD* encodes a protein of unknown function. The ABC transport apparatus is encoded by *rbsACB*. *rbsK* encodes the cytoplasmic ribokinase, which phosphorylates ribose. *rbsR* encodes a repressor protein that regulates transcription of the *rbs* operon.
In both (A) and (B), black boxes represent promoters, and thin arrows indicate the direction of transcription.
C. A schematic of the ribose transport system.
Similar shading patterns indicate homologous genes in (A) and (B), and protein functions in (C).

in the *S*-adenosylmethionine (SAM) utilization pathway. Specifically, LuxS cleaves SRH to form homocysteine and AI-2 (Schauder *et al.*, 2001; Schauder and Bassler, 2001). The current evidence suggests that, in contrast to the variable structures of acyl-HSL and peptide autoinducers, the structures of AI-2 molecules from different species of bacteria are identical. If AI-2 is used for interspecies signalling in natural habitats, a common signal structure could be required in order for it to be recognized by multiple members of a mixed-species community (Schauder *et al.*, 2001; Schauder and Bassler, 2001).

Although the role of AI-2 is understood in the regulation of bioluminescence in *V. harveyi*, what function AI-2 plays, if any, in other *luxS*-containing bacteria is not clear. There are reports showing that AI-2 is involved in regulating type III secretion in *Escherichia coli* O157:H7 (Sperandio *et al.*, 1999), protease production in *Streptococcus pyogenes* (Lyon *et al.*, 2001), haemolysin production in *Vibrio vulnificus* (Kim *et al.*, 2000) and regulation of the virulence factor VirB in *Shigella flexneri* (Day and Maurelli, 2001). In the present report, we have performed a series of genetic experiments in an attempt to determine how AI-2 regulates gene expression in *Salmonella typhimurium*. We show that AI-2 controls the expression of a previously uncharacterized operon encoding an ABC transporter apparatus that appears to function in the uptake of AI-2.

Results

Identification of genes regulated by luxS *in* S. typhimurium

We used the following strategy to identify *luxS*-regulated target genes in *S. typhimurium*. MudJ transposon mutagenesis was used to generate isogenic *lacZ* transcriptional fusions in wild-type and *luxS* null strains of *S. typhimurium* (Hughes and Roth, 1988; see *Experimental procedures* for details). A total of 11 000 insertion mutants was made and subsequently screened for differential expression of the *lacZ* reporter gene. The MudJ insertions from candidate *luxS*-regulated fusion strains were back-crossed by P22 transduction into *S. typhimurium* 14028 (wild type) and *S. typhimurium* SS007 (*luxS*::T-POP) to verify that the differences we observed in *lacZ* expression were a consequence of the presence or absence of a functional *luxS* gene on the *S. typhimurium* chromosome. We identified eight *lacZ* fusions that appeared to produce higher levels of β-galactosidase in the strain carrying the wild-type *luxS* gene than in the *luxS* null strain.

The insertion sites of the eight *luxS*-regulated MudJ fusions were determined by arbitrary polymerase chain reaction (PCR) amplification of the MudJ–chromosome fusion junctions, followed by BLAST database analysis (Altschul *et al.*, 1990; Caetano-Anolles, 1993). One MudJ insertion was located in the *metE* gene. MetE is involved in the biosynthesis of methionine, which is a precursor of SAM (Weissbach and Brot, 1991). Specifically, MetE catalyses the final step in methionine biosynthesis in the absence of vitamin B12. Transcription of *metE* is induced when the substrate homocysteine is available and when the intracellular concentration of methionine is low (Urbanowski and Stauffer, 1989). We reason that *metE* is not a true target of AI-2 regulation but, instead, *metE* transcription is induced in the wild-type *luxS* strain relative to the *luxS* null strain because homocysteine is produced during the generation of AI-2 in the LuxS^{+} strain.

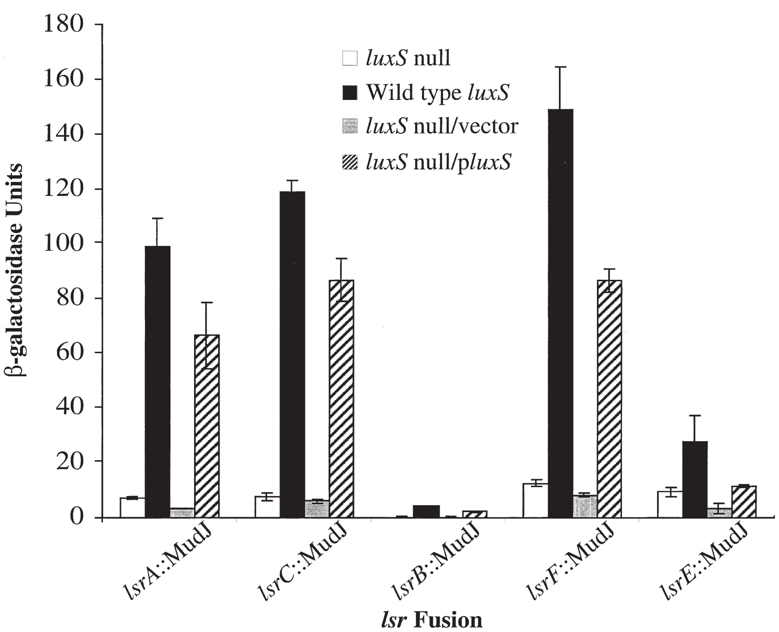

Fig. 2. *luxS* regulates the transcription of the *lsr* operon. The β-galactosidase activity is shown for five MudJ–*lacZ* reporter fusions in genes in the *lsr* operon. The white and black bars show the activity for the *lsr*::MudJ fusions in *luxS*::T-POP (denoted *luxS* null) and wild-type *luxS* backgrounds respectively. The grey and striped bars show the activity when the plasmids pUC18 (parental vector) and pAB15 (pUC18 expressing wild-type *luxS*) were introduced into the *luxS*::T-POP strains respectively.

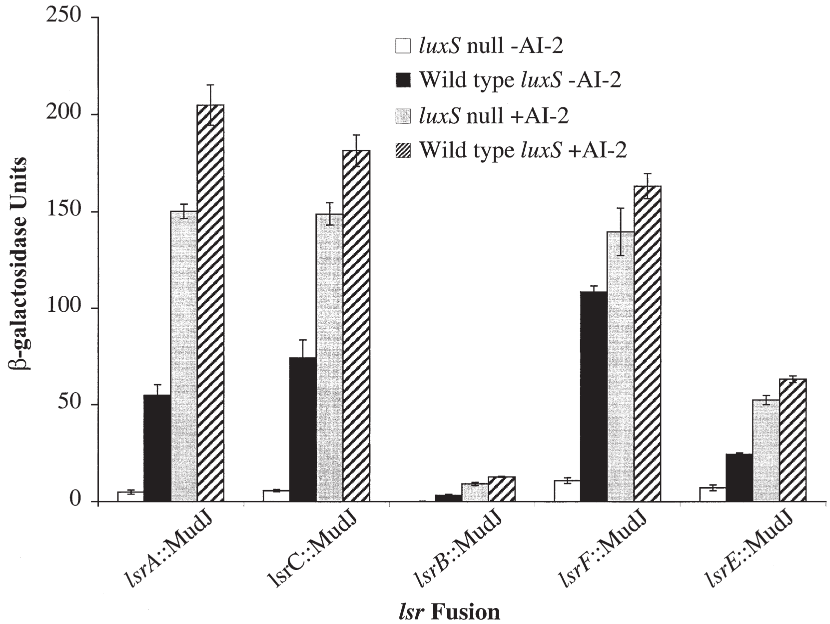

Fig. 3. AI-2 regulates *lsr* operon expression. The β-galactosidase activity of strains carrying MudJ fusions in five *lsr* genes was assayed in *luxS*::T-POP (denoted *luxS* null) and wild-type *luxS* backgrounds. Cultures were grown in LB containing either a control reaction mixture (denoted −AI-2) or *in vitro*-synthesized AI-2 (denoted +AI-2). *In vitro* AI-2 was prepared by incubating *S*-adenosylhomocysteine and the purified Pfs and LuxS proteins as described previously (Schauder *et al.*, 2001). The control mixture (−AI-2) was prepared by an identical procedure except that the reaction was carried out in the absence of LuxS protein. (D,L)-homocysteine was added to this control preparation after the reaction to compensate for the homocysteine produced in the reaction containing LuxS. AI-2 was used at an estimated concentration of 70 μM in this experiment.

Consistent with this hypothesis, we found that the addition of exogenous AI-2 to the *metE–lacZ* insertion mutant did not affect the transcription of the *metE–lacZ* fusion (not shown). We did not study the *metE–lacZ* fusion strain further.

The seven remaining MudJ insertions all resided in a single operon in *S. typhimurium* (Fig. 1A). This operon is of unknown function but is predicted to encode an ABC transporter complex with striking similarity to the ribose transport operon (Rbs) of *E. coli* and *S. typhimurium*. We have named this operon the *lsr* operon for *luxS* regulated. Figure 1A shows the *lsr* operon of *S. typhimurium* and the sites of the *luxS*-regulated MudJ insertions that we identified in our genetic screen.

The *rbs* operon of *E. coli* and *S. typhimurium* is shown in Fig. 1B. The functions encoded by the *rbs* operon are responsible for the high-affinity transport and phosphorylation of ribose (Fig. 1C; Iida *et al.*, 1984). The *rbs* operon has been studied most extensively in *E. coli*, although the *rbs* operon of *S. typhimurium* has the same arrangement and is predicted to encode proteins with identical functions. In the *rbs* operon of *E. coli*, the first gene, *rbsD*, encodes a protein with no known function. RbsA, RbsB and RbsC comprise the transport apparatus (Bell *et al.*, 1986). RbsB is the periplasmic-binding protein that recognizes ribose and interacts with RbsC. RbsC is the homodimeric channel-forming protein (Park *et al.*, 1999). RbsA is the membrane-associated ATPase that supplies energy to drive the transport of the sugar into the cell (Buckel *et al.*, 1986). RbsK is a cytoplasmic kinase that phosphorylates ribose upon entry into the cell (Hope *et al.*, 1986). Finally, RbsR is the regulatory protein that functions at the *rbs* promoter to repress transcription of the operon in the absence of ribose (Mauzy and Hermodson, 1992).

In the *lsr* operon, the first gene, *lsrA*, encodes a protein homologous to the RbsA ATPase. The *lsrC* and *lsrD* genes are homologous to one another and to *rbsC* and are predicted to encode the components of a heterodimeric membrane channel. *lsrB* encodes a predicted sugar-binding protein homologous to RbsB. In the *lsr* operon, three additional genes are located downstream of the putative ABC transporter. We have named these genes *lsrF*, *lsrG* and *lsrE*. The *lsrF* and *lsrG* genes encode proteins of unknown function, and *lsrE* encodes a protein with homology to the *E. coli* ribulose phosphate epimerase (*rpe*; Sprenger, 1995). The completed genome sequence of *E. coli* shows that it also possesses this operon (Blattner *et al.*, 1997). In the *E. coli* genome, the operon has been designated the *b1513* operon. The *b1513* operon has an arrangement identical to that of the *S. typhimurium lsr* operon, except that the *E. coli b1513* operon does not contain a gene homologous to *lsrE*.

Analysis of the luxS*-regulated fusions in the* lsr *operon*

The β-galactosidase activity produced by each of the *lacZ* fusions in the *lsr* operon was measured in wild-type *luxS and luxS* null *S. typhimurium* strains. The results are shown in Fig. 2. Note that, although we obtained three insertions in the *lsrF* gene, results for only one of these fusions appear in Fig. 2, because all three fusions have the same activity. The white bars show the β-galactosidase activity in the *luxS* null strain, and the black bars show the corresponding level of β-galactosidase activity for each fusion when wild-type *luxS* is present. The figure shows that the *lsrA*, *lsrC* and *lsrF* fusions are induced 12- to 16-fold when wild-type *luxS* is present.

The *lsrB–lacZ* and *lsrE–lacZ* fusions show lower activity than the fusions we obtained in the other three genes in this operon. In the case of *lsrE*, as this is the gene most distal to the promoter, the low activity may be explained by weak transcription. However, weak transcription cannot be the cause of the low activity observed for the *lsrB–lacZ* fusion, because we observe high activity from the downstream *lsrF–lacZ* fusion. One explanation for the low activity in the *lsrB–lacZ* fusion is that a functional LsrB protein is required for the transcription of the *lsr* operon. To test this hypothesis, we constructed an in frame deletion of *lsrB* and demonstrated that it did not affect the transcription of any of the other *lsr–lacZ* fusions (not shown). Furthermore, although the absolute β-galactosidase units are low for the *lsrB–lacZ* fusion, the fold induction of this fusion in the wild-type *luxS* strain over the *luxS* null strain is about the same as that of all the other fusions we obtained in *lsr* genes. Based on these results, we conclude that this particular fusion joint is uncommonly polar and results in lower than expected transcription of *lacZ*, which would account for its reduced β-galactosidase activity.

To verify that the induction of transcription of the *lsr* operon results from *luxS*, we complemented the fusion strains with the *luxS* gene *in trans*. Figure 2 shows these data. In a *luxS* null *S. typhimurium* background, introduction of the parent vector without *luxS* had no effect on the β-galactosidase expression of the *lsr* fusions (grey bars). However, introduction of the cloned *luxS* gene into each of the *lsr–lacZ* fusion strains restored β-galactosidase activity to nearly the level observed for the fusions in the wild-type *luxS* background (striped bars).

AI-2 regulates the transcription of the lsr *operon*

The above results show that the *lsr* operon is induced in the presence of a functional *luxS* gene. In earlier work, we have shown that the LuxS enzyme catalyses the final step in the biosynthesis of AI-2 (Schauder *et al.*, 2001). We interpret our results above to mean that regulation of the *lsr* operon by *luxS* is mediated by the presence or absence of the AI-2 signal molecule. We further predict that AI-2 should exert its regulatory effect on the *lsr* operon by signalling from the outside of the cell. However, the strategy we used to identify the *lsr* operon relies on internal production of AI-2 in *S. typhimurium*. To prove that the AI-2 molecule is sufficient for *lsr* regulation, and also to prove that external AI-2 is capable of mediating this effect, we tested whether exogenously supplied AI-2 could induce the *lsr–lacZ* fusions in *S. typhimurium* strains containing a null mutation in *luxS*.

Recently, we developed a procedure for the *in vitro* production of *S. typhimurium* AI-2 from S-adenosylhomocysteine (SAH) using purified Pfs and LuxS proteins (Schauder *et al.*, 2001). We can estimate the AI-2 concentration in our *in vitro* preparations indirectly by measuring the concentration of homocysteine that is produced along with AI-2 by LuxS. Using AI-2 prepared by this method, we tested the effect of the exogenous addition of AI-2 on the regulation of the *S. typhimurium lsr* operon.

Various *S. typhimurium lsr*::MudJ fusion strains were

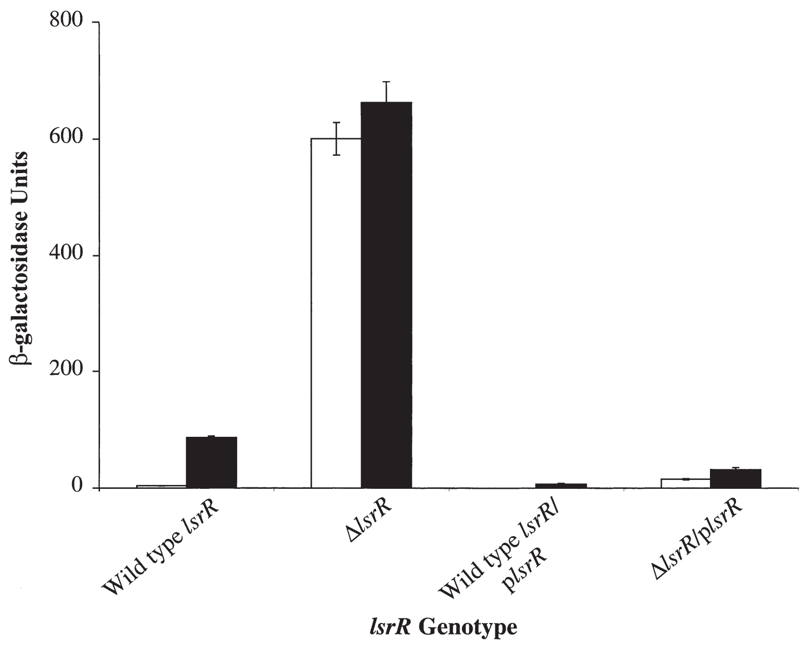

Fig. 4. The LsrR protein is required for AI-2 control of *lsr* operon expression. β-galactosidase activity was measured from the *lsrC*::MudJ reporter in wild-type *lsrR* and Δ*lsrR* *S. typhimurium* strains. Additionally, the β-galactosidase activity was measured in these same strains after the introduction of the vector pMET1035, which carries the wild-type *lsrR* gene (denoted p*lsrR*). The white bars show the activities in a *luxS* null *S. typhimurium* strain background, and the black bars show the activities in a wild-type *luxS S. typhimurium* strain background.

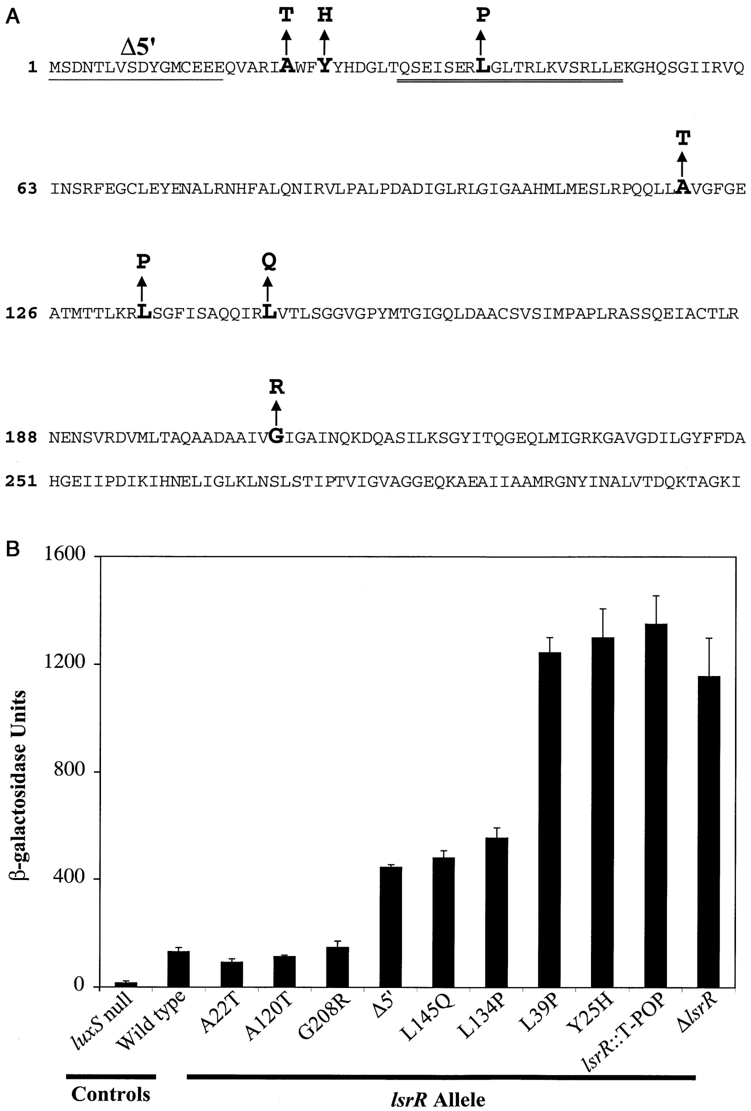

Fig. 5. Activity of LsrR suppressor mutants. Eight spontaneous mutations in *lsrR* were identified that conferred a Lac^+ phenotype to an *S. typhimurium lsrC*::MudJ, *luxS* null strain.
A. The LsrR protein sequence and the amino acid alterations causing a suppressor phenotype. The spontaneous deletion that resulted in a suppressor phenotype is shown by a single underline (denoted Δ5′). The predicted helix–turn–helix DNA-binding motif is shown by the double underline.
B. The corresponding β-galactosidase activity is shown for the *lsrC*::MudJ reporter in each of the eight suppressor strains and the activity for one of the *lsrR*::T-POP suppressors we obtained. In every case, the chromosomal copy of the *luxS* gene has been inactivated by insertion. As a reference, the β-galactosidase activity from the *lsrC*::MudJ reporter is also shown for *S. typhimurium* strains containing the wild-type *lsrR* gene (controls) and an in frame deletion of *lsrR* (denoted Δ*lsrR*).

grown to late exponential phase in LB medium supplemented with AI-2 prepared by our *in vitro* method at an approximate concentration of 70 μM. This is roughly twice the concentration of AI-2 that we estimate to be present in *S. typhimurium* cell-free supernatants (not shown). As a control, we performed the same experiment with material from *in vitro* reactions carried out with SAH and Pfs enzyme in the absence of the LuxS enzyme. This reaction produces adenine and SRH. Homocysteine was added to this control mixture after the reaction to compensate for the homocysteine that would be produced by LuxS. Therefore, the only component absent from the control reaction was AI-2. The β-galactosidase measurements in Fig. 3 show that, in the presence of the −AI-2 control reaction, low expression of the *lsr–lacZ* fusions occurs in the absence of *luxS* (white bars) and, similar to what was shown in Fig. 2, the fusions are induced to varying degrees in the presence of wild-type *luxS* (black bars). In strains lacking *luxS*, the addition of 70 μM exogenous AI-2 induces the expression of the *lsr–lacZ* fusions to levels above those when wild-type *luxS* is present on the chromosome (grey bars). This experiment shows that exogenous AI-2 is sufficient for induction of the *lsr* operon in *S. typhimurium*.

We also supplied exogenous AI-2 to the *S. typhimurium lsr–lacZ* fusion strains that were wild type for *luxS* on the chromosome (striped bars). Figure 3 shows that increasing the level of AI-2 above the endogenous level produced by chromosomal *luxS* results in an additional stimulation of the expression of the *lsr–lacZ* fusions. This result indicates that, under the conditions in which we are performing our experiments, the *lsr* operon is not fully induced by the endogenously produced AI-2.

Identification of LsrR: a protein responsible for mediating AI-2 regulation of transcription of the lsr *operon*

The above results show that AI-2 induces the expression of the *lsr* operon. However, because no DNA-binding protein is encoded in the *lsr* operon, we did not understand how the presence of AI-2 could modulate the transcription of the *lsr* operon. We hypothesized that a DNA-binding protein must exist in *S. typhimurium* that functions to couple the presence of the AI-2 signal to the expression of the *lsr* operon. To identify this regulatory factor, we performed a genetic selection for mutants in which the Lac$^-$ phenotype of a *luxS* null, *lsr–lacZ* fusion strain was suppressed. Our reasoning was as follows: the *luxS* null strains containing *lacZ* fusions in the *lsr* operon do not grow on lactose minimal plates. However, the presence of a functional *luxS* gene on the chromosome induces the expression of the *lacZ* gene in these fusions to a level sufficient to restore growth on lactose minimal plates. Therefore, selection for growth of a *luxS* null, *lsr–lacZ S. typhimurium* strain on lactose minimal plates enabled us to isolate *S. typhimurium* colonies that had acquired spontaneous suppressor mutations that allowed increased expression of the *lsr–lacZ* fusion in the absence of *luxS* (i.e. in the absence of AI-2). We performed this experiment with the representative *lsrC–lacZ* strain. Eight spontaneous Lac$^+$ colonies were isolated by this technique and studied further. We also used a transposon mutagenesis strategy to identify the putative regulatory factor. Mutagenesis with the transposon T-POP was performed on the *luxS* null, *lsrC–lacZ* fusion strain. Approximately 12 000 insertions were made, and two transposon insertion mutants were identified that restored growth to the *luxS* null, *lsrC–lacZ* strain on lactose minimal medium.

Linkage analysis showed that the T-POP insertions and the spontaneous suppressor mutations could be co-transduced with high frequency (76% co-transduction) with the *lsrC*::MudJ insertion. Database analysis of the *S. typhimurium* genome revealed that a gene encoding a predicted DNA-binding protein of unknown function is located immediately upstream of the *lsr* operon but is transcribed divergently (Fig. 1). This protein is homologous to SorC, a DNA-binding protein that functions as both an activator and a repressor of transcription of genes involved in sorbose metabolism in *Klebsiella pneumoniae* (Wehmeier and Lengeler, 1994). We predicted that the transposon and spontaneous suppressor mutations could be located in this gene. Indeed, we were able to PCR amplify the chromosome–T-POP fusion junction from both T-POP suppressor strains using primers specific to T-POP and the regions flanking this putative regulatory gene. In addition, we PCR amplified and sequenced this regulatory gene from the chromosome of the eight spontaneous suppressor strains. In every case, this gene was found to contain either a point mutation or a deletion. We have named this gene *lsrR*, for regulator of the *lsr* operon. Because inactivation of the *lsrR* gene results in high-level expression of the *lsr* operon in a *luxS* null background, these results strongly suggest that the wild-type function of LsrR is to repress the expression of the *lsr* operon in the absence of AI-2.

Deletion of lsrR *results in unregulated transcription of the* lsr *operon*

To verify the role of LsrR in AI-2 regulation of *lsr* operon transcription, we constructed an in frame deletion of *lsrR* in *S. typhimurium* and assayed the expression of the *lsr–lacZ* fusions in the presence and absence of *lsrR*. The results are shown in Fig. 4 for the representative *lsrC–lacZ* reporter fusion. The first pair of bars shows that, as in Fig. 2, when wild-type *lsrR* is present, *lsrC–lacZ* expression is repressed in a *luxS* null strain and derepressed 23-fold when *luxS* is present and the AI-2 signal molecule is synthesized (white and black bars respectively). In contrast, the second pair of bars shows that deletion of *lsrR* results in high-level expression (over 100-fold derepression) of the *lsrC–lacZ* reporter whether or not *luxS* is present. This result demonstrates that the *lsr* operon cannot be regulated by AI-2 in the absence of LsrR. We cloned the wild-type *lsrR* gene and expressed it *in trans* under an exogenous promoter to show that LsrR could complement the Δ*lsrR* defect. The third and fourth pairs of bars in Fig. 4 show that *in trans* expression of *lsrR* in both wild-type *lsrR* and Δ*lsrR S. typhimurium* strains causes repression of the *lsrC–lacZ* reporter fusion. Specifically, complete repression of *lsrC–lacZ* transcription occurs in the *luxS* null strains (white bars), and nearly complete repression of *lsrC–lacZ* transcription occurs in the wild-type *luxS* strains (black bars). These results show that the introduction of *lsrR* complements the Δ*lsrR* defect. However, when *lsrR* is overexpressed in a wild-type *luxS* background, the AI-2 produced is apparently insufficient to compensate for the increased LsrR protein produced by the plasmid-encoded gene. This imbalance results in incomplete inactivation of LsrR by AI-2, so *lsr* operon repression occurs. We performed control experiments to show that the parent vector without the cloned *lsrR* gene had no effect on *lsrC–lacZ* activity in any of these strains (not shown).

Analysis of the LsrR suppressor mutations

The sequence of the LsrR protein is shown in Fig. 5A as well as the spontaneous mutations we identified in the suppressor selection. The β-galactosidase activities of the spontaneous *lsrR* mutations were measured and compared with those of wild-type *lsrR* and the *lsrR* deletion.

Again, the *lsrC–lacZ* fusion is used as the representative reporter for the *lsr* operon. Our results are shown in Fig. 5B. The two leftmost bars show the control experiments in which *lsrC–lacZ* expression is measured in the *luxS* null and wild-type *luxS* strains. Both these strains contain wild-type *lsrR*. As shown above, expression of *lsrC–lacZ* is derepressed in the wild-type *luxS* strain compared with the *luxS* null strain (in this case ninefold). The remaining bars in Fig. 5B represent the activities of various *lsrR* mutants in a *luxS* null background. The spontaneous mutations resulted in derepression of the *lsrC–lacZ* fusion from sixfold to a maximum of 250-fold (L39P and Y25H mutants). This high level of derepression caused by the L39P and Y25H mutants represents maximal derepression of the operon, because we observe this same level of derepression in the null *lsrR*::T-POP and Δ*lsrR* mutations. Note that, although we obtained two T-POP insertions in *lsrR*, the activity for only one of the insertions is shown. Both insertions gave identical results in this experiment.

The Lsr ABC transporter has a role in the elimination of extracellular AI-2

The *lsr* operon apparently encodes an ABC transporter resembling the ribose transporter. As AI-2 is a ribose derivative, we hypothesize that AI-2 could be the ligand for the Lsr transporter complex. Therefore, one possible function of the Lsr operon is to transport the AI-2 synthesized by LuxS in the cytoplasm out of the

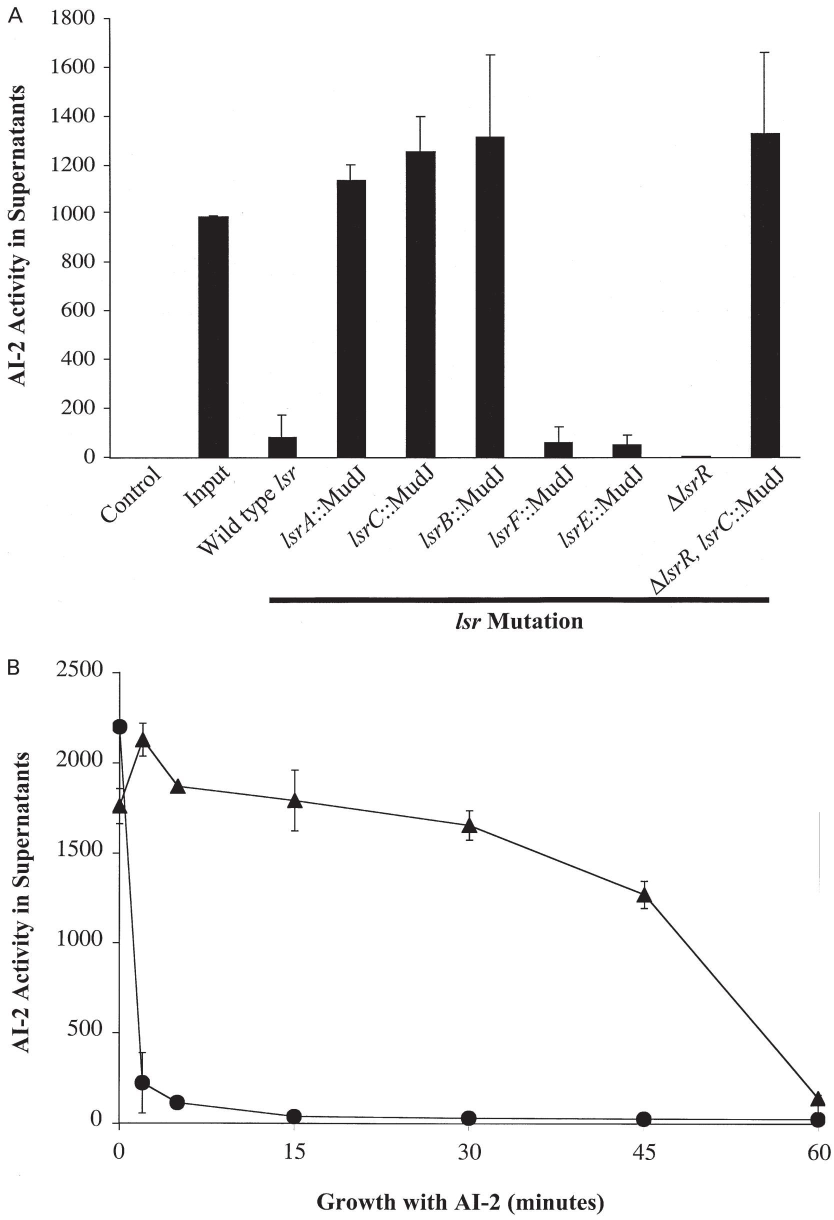

Fig. 6. The Lsr ABC transporter is required for removal of AI-2 from *S. typhimurium* culture fluids. The quantity of AI-2 remaining in cell-free culture fluids of *S. typhimurium* strains containing various mutations in *lsr* genes is shown. AI-2 activity in the culture fluids was measured using the *V. harveyi* BB170 AI-2 bioassay. All strains tested contained a null mutation in *luxS*, so the only AI-2 present was that added exogenously.
A. Various *S. typhimurium* strains were grown for 4 h. *In vitro*-prepared AI-2 was included in the cultures during the final hour of growth. The control bar represents the AI-2 activity for the culture fluid of the *S. typhimurium luxS* null parent strain SS007 (*luxS*::T-POP) when no AI-2 was added. The input bar represents the amount of AI-2 activity added to each culture. The rest of the bars show the AI-2 activity remaining in the culture fluids of the wild-type *lsr* strain and the various *lsr* mutants after the 1 h incubation period.
B. Time courses of elimination of AI-2 from the culture fluids are shown for the wild-type *lsrR* strain SS007 (*luxS*::T-POP, triangles) and the Δ*lsrR* strain MET342 (Δ*lsrR*, *luxS*::T-POP, circles). These cultures were grown for a total of 4 h. During the final hour of growth, AI-2 was included in the cultures for the times specified on the *x*-axis.

S. typhimurium cell. This possibility seems unlikely because the *lsr* operon encodes a transporter with homology to binding protein-dependent ABC transporters that function to import molecules into the cytoplasm (Nikaido and Hall, 1998). Moreover, mutants with insertions in the *lsr* operon are not deficient in AI-2 production, indicating that the Lsr transporter is not required for export of the AI-2 signal into the medium (not shown).

We have reported previously that, in *S. typhimurium*, extracellular AI-2 activity accumulates to maximal levels in late exponential phase and, subsequently, the AI-2 activity disappears from the medium (Surette and Bassler, 1998; 1999). Additionally, we have shown that AI-2 in cell-free culture fluids remains active for long periods of time, indicating that the disappearance of the activity from the extracellular environment is not the result of instability of the AI-2 molecule (Surette and Bassler, 1999). Furthermore, we have shown that elimination of AI-2 from the medium requires protein synthesis (Surette and Bassler, 1999). Therefore, an alternative possibility for the function of the Lsr complex is to import extracellular AI-2 into the cytoplasm of *S. typhimurium*. We performed an experiment to investigate this possibility.

Cultures of *S. typhimurium luxS* null strains containing a wild-type *lsr* operon or containing mutations inactivating *lsr* genes were grown to mid-exponential phase in LB at 37°C. AI-2 prepared by our *in vitro* method was added to the cultures, and the cultures were incubated with the AI-2 for 1 h. Subsequently, the cells were removed from the culture fluid by centrifugation, and the resulting cell-free culture fluids were assayed for AI-2 activity using the *V. harveyi* AI-2 bioassay. This assay involves measuring the increase in light production of a *V. harveyi* quorum-sensing reporter strain that induces bioluminescence exclusively in response to the presence of AI-2. In the present experiment, the bioassay allowed us to determine the level of AI-2 activity remaining in the culture fluids of the different *S. typhimurium* mutants. As all the *S. typhimurium* strains we used in this experiment contained null mutations in *luxS*, the only AI-2 that could be present in the cell-free culture fluids came from that which we added exogenously.

Figure 6A shows the results of this experiment. The control experiment shows that, as expected, no AI-2 activity is present in culture fluids prepared from the *S. typhimurium luxS* null strain that we used as the parent for these studies. The bar labelled 'Input' shows that AI-2 sufficient to induce the *V. harveyi* bioluminescence reporter 1000-fold was added to the remainder of the cultures. The input activity was measured by adding AI-2 to the parent *S. typhimurium* culture and immediately chilling the culture on ice, followed by centrifugation of the cells and isolation of the cell-free culture fluid. Figure 6A shows that a 1 h incubation of the wild-type *lsr* strain with the input AI-2 results in the disappearance of 92% of the AI-2 activity from the culture fluid, as activity sufficient to induce the bioassay strain only 80-fold remained (see the bar labelled wild-type *lsr*). In contrast, *S. typhimurium* strains containing MudJ insertion mutations in *lsrA*, *lsrC* and *lsrB* were unable to eliminate AI-2 from the culture fluids, as AI-2 activity equivalent to the input activity remained in the culture fluids after the incubation period. In addition, a strain containing an in frame deletion in *lsrB* is also unable to eliminate AI-2 from the supernatant in this assay (not shown). These results indicate that a functional Lsr ABC transporter is required for the process of removal of AI-2 from the medium.

MudJ insertions in *lsrF* and *lsrE* did not impair the ability of *S. typhimurium* to remove AI-2 from the medium. Figure 6A shows that, similar to when the wild-type *lsr* operon is present, 95% of the input activity disappears after a 1 h incubation in the *lsrF* and *lsrE* mutants. The *lsrF* and *lsrE* genes are located downstream of the genes encoding the structural components of the transport apparatus (Fig. 1). Therefore, although the MudJ insertions generating *lacZ* transcriptional fusions to *lsrE* and *lsrF* are induced by AI-2 by virtue of their presence in the *lsr* operon, LsrE and LsrF are not expected to be required for building a functional transport apparatus. Taken together, our results indicate that only the genes that encode the structural components of the Lsr transporter are required for removal of AI-2 from *S. typhimurium* cell-free culture fluids.

Figure 6A also shows that inactivation of *lsrR* results in the highest level of removal of AI-2 from the extracellular culture fluids of *S. typhimurium*, as over 99% of the activity is gone after the 1 h incubation. To verify that elimination of AI-2 from the medium that we observed in this experiment is dependent on the Lsr transporter, we combined the Δ*lsrR* mutation with the *lsrC*::MudJ insertion and performed an identical experiment. The final bar in Fig. 6A shows that, regardless of the presence or absence of *lsrR*, inactivation of the Lsr transporter renders *S. typhimurium* unable to eliminate AI-2 from the medium because, in the Δ*lsrR*, *lsrC*::MudJ double mutant, the level of AI-2 did not decrease compared with the input activity.

Deletion of *lsrR* results in maximal removal of the AI-2 from the culture fluids. This result is expected because the Δ*lsrR* strain displays completely derepressed expression of the *lsr* operon, and this strain would therefore be predicted to have increased levels of the Lsr transporter compared with the wild-type strain. If this is the case, the Δ*lsrR* strain should have an enhanced capability to remove AI-2 from the medium. To test this idea, we assayed the rate of disappearance of AI-2 from the medium over time for the Δ*lsrR* mutant and compared that with the rate of disappearance for the wild-type strain. The results are presented in Fig. 6B. This experiment shows that removal of AI-2 from the medium by the wild-type *lsrR* strain occurs rather steadily over the first 45 min of the incubation period, then more rapidly between 45 min and 1 h (triangles). In contrast, in the Δ*lsrR* mutant, only 10% of the AI-2 activity remains after 2 min, and all but 2% of the activity is gone by 15 min (circles). This result demonstrates that the derepression of the expression of the Lsr transport complex increases the efficiency of elimination of AI-2 from the external environment of *S. typhimurium*.

Discussion

In our previous investigations of the model luminous bacterium *V. harveyi*, we identified a quorum-sensing signal molecule that we called AI-2 and showed that it is involved in induction of the expression of luciferase (Bassler *et al*., 1993; 1994a). Subsequently, we showed that AI-2 and the synthase required for its production, LuxS, are present in a wide variety of bacterial species (Bassler *et al*., 1997; Bassler, 1999; Surette *et al*., 1999; Miller and Bassler, 2001). We have suggested that AI-2 is used for interspecies communication among bacteria (Bassler *et al*., 1997; Surette *et al*., 1999; Miller and Bassler, 2001). However, it is not clear how AI-2 is detected nor what functions are regulated by this signal in most species of bacteria. In the present report, we have performed experiments aimed at identifying the AI-2-regulated genes and the mechanism of AI-2 signal transduction in the enteric pathogen *S. typhimurium*.

In a genetic screen for functions regulated by AI-2 in *S. typhimurium*, we identified an operon predicted to encode an ABC transporter with similarity to the ribose transporter of *E. coli* and *S. typhimurium* (Fig. 1). This operon (the *lsr* operon) is induced in the presence of *luxS* and AI-2 (Figs 2 and 3), and AI-2 regulation of *lsr* operon expression requires the LsrR DNA-binding regulatory protein (Figs 4 and 5). We suggest that the function of the Lsr transporter is to import AI-2 from the external environment because strains possessing mutations in the Lsr transporter do not remove AI-2 from the culture medium, whereas AI-2 disappears from culture fluids of *S. typhimurium* strains possessing a wild-type Lsr transporter. Additionally, strains derepressed for *lsr* expression by inactivation of *lsrR* remove the AI-2 more rapidly than wild type (Fig. 6). In earlier work, we showed that accumulation of AI-2 by *S. typhimurium* is maximal in late exponential phase, and all the AI-2 activity disappears from the medium by the time stationary phase is reached (Surette and Bassler, 1998). We propose that the Lsr transporter is involved in the AI-2 removal process.

In Fig. 7, we present a model for the putative function of the Lsr transporter and its regulation by AI-2 and LsrR. LuxS synthesizes the AI-2 signal molecule in the cytoplasm. AI-2 is released to the external environment by an unknown mechanism. Extracellular AI-2 signals to

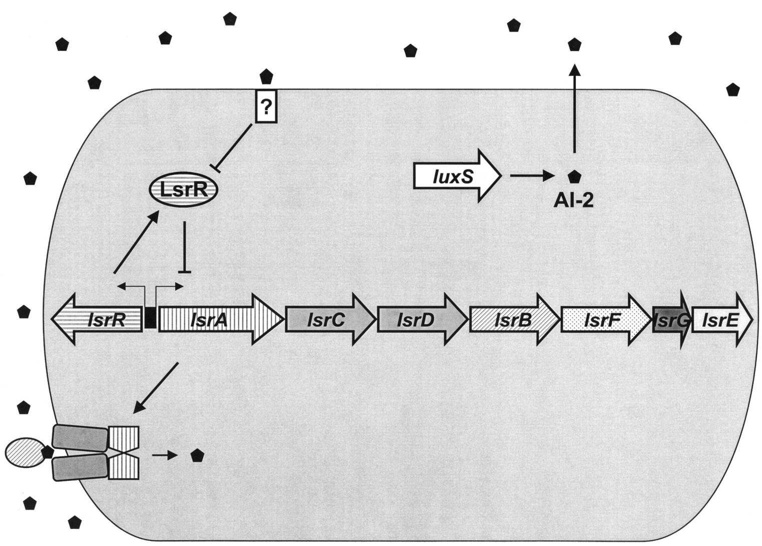

Fig. 7. A model for the regulation and function of the Lsr ABC transporter in *S. typhimurium*. In the absence of AI-2, the LsrR DNA-binding protein represses the transcription of the *lsr* operon. LuxS synthesizes AI-2 (black pentagons), and AI-2 is released from the cell by an unknown mechanism. External AI-2 signals to the LsrR protein. This step inactivates the LsrR repressor function, which promotes transcription of the *lsr* operon. The mechanism of AI-2 signalling to LsrR is not known, but we propose that a sensor exists that couples AI-2 detection to LsrR activity. The white box containing the question mark represents the putative sensor. Transcription of the *lsr* operon results in the production of the Lsr ABC transporter complex, which functions to import AI-2 into *S. typhimurium*. The LsrACBD components are shown as the complex in the bacterial membrane. We hypothesize that AI-2 interacts with the periplasmic LsrB protein (represented by the stippled oval), and that LsrB–AI-2 binding facilitates interaction with the channel and ATPase components of the Lsr complex for transport. Note that the ABC transporter is predicted to reside in the inner membrane, with LsrB in the periplasm (grey and striped boxes). We have not drawn the outer membrane of the bacterial cell.

the LsrR protein, again by an unknown mechanism. In Fig. 7, we suggest that a sensor exists that is responsible for the detection of extracellular AI-2 and signal transduction to LsrR. Other possibilities exist and are discussed below. In the absence of AI-2, LsrR acts to repress transcription of the *lsr* operon, most probably by binding at the *lsr* operon promoter analogous to RbsR binding at the *rbs* promoter (Mauzy and Hermodson, 1992). AI-2 signalling to LsrR alleviates LsrR repression of the *lsr* operon, resulting in transcription of the genes encoding the Lsr ABC transport apparatus. The Lsr transporter, in turn, imports AI-2 from the extracellular environment into the cells. We do not know what becomes of the AI-2 after import.

Two main types of ABC transporters exist in bacteria. One type of ABC transporters consists of binding protein-dependent transporters that function to import small molecule ligands into the cell (Boos and Lucht, 1996; Nikaido and Hall, 1998; Holland and Blight, 1999). The Lsr transporter has the highest homology to this class of ABC transporters, whose members include the RbsACB ribose transporter, the MglBAC galactose transporter and the HisJQMP histidine transporter (Boos and Lucht, 1996). The second class of ABC transporters functions to export molecules out of the cell. These transporters do not require a periplasmic binding protein, and their ligands vary greatly in size and composition (Boos and Lucht, 1996). The LmrA multidrug resistance transporter, the HlyB haemolysin transporter of *E. coli* and the ComAB and SapTE oligopeptide autoinducer transporters of *Streptococcus pneumoniae* and *Lactobacillus sake* are examples of this second type of ABC exporter (Kleerebezem *et al*., 1997; Holland and Blight, 1999). Additionally, ABC transporters are used for the transport of molecules in higher organisms, including humans. Deficiency in the function of ABC transporters has been implicated in many human diseases, the most well-studied example being the CFTR chloride ion transporter that is non-functional in cystic fibrosis patients (Holland and Blight, 1999).

The homology between the Lsr complex and the family of ABC transporters suggests a role for the Lsr complex in the transport of a ligand. Furthermore, because AI-2 is a ribose derivative, because the Lsr transporter most closely resembles the ribose transporter and because our evidence shows that the *lsr* genes are required for the removal of AI-2 from the external environment, we suggest that AI-2 is the ligand for the Lsr transporter. These findings, coupled with the facts that the Lsr transporter most closely resembles the importer type of ABC transporter and that the Lsr complex is not required for the export of AI-2, led us to hypothesize that the Lsr complex transports AI-2 into the cell. In *V. harveyi*, the LuxP protein is the primary receptor for AI-2. The LuxP–AI-2 complex initiates the quorum-sensing signalling cascade (Bassler *et al*., 1994a). LuxP is a periplasmic protein homologous to the ribose-binding protein RbsB and to LsrB, the ribose-binding protein homologue we identified in the present work. We suggest that LsrB is the AI-2-binding protein in *S. typhimurium*. In this case, instead of initiating a signalling cascade, the function of LsrB is to bring AI-2 into contact with the membrane-bound components of the Lsr apparatus for transport into the cell. However, until we prove this biochemically, it remains possible that the Lsr complex is required for the modification or degradation of AI-2 on the cell surface.

We have shown previously that a secreted enzyme is not responsible for the elimination of AI-2 from the medium because AI-2 is stable for long periods of time in *S. typhimurium* cell-free spent culture fluids (Surette and Bassler, 1999). We have also observed that AI-2 does not disappear from the supernatants of dead cells (not shown). These results together indicate that the elimination of AI-2 occurs either at the surface of living cells or, more likely, based on the findings in this report, is mediated by its transport into the cell via the Lsr complex. However, at least one other Lsr-independent mechanism for AI-2 elimination must exist in *S. typhimurium* because, in both wild-type and *lsrA*, *lsrB* and *lsrC* mutant strains, although AI-2 activity does not decrease during the first hour of incubation, the AI-2 activity decreases in culture fluids after longer periods of incubation (20 h; data not shown).

The mechanism of AI-2 signalling to LsrR to derepress transcription of the *lsr* operon can be from the inside or the outside of the cell. If AI-2 acts from the inside, then AI-2 must enter the cells by some mechanism that is not dependent on the Lsr transporter. We conclude this because *S. typhimurium* strains harbouring mutations in the *lsr* operon, although deficient in the process of internalization of AI-2, remain capable of AI-2-dependent derepression of transcription of the *lsr* operon. Therefore, if AI-2 is required to enter the cell to signal to LsrR, it does not require the Lsr apparatus and, furthermore, only a minor amount of AI-2 internalization can be required for the signalling process. We say this because we cannot detect any disappearance of AI-2 from culture fluids prepared from strains containing mutations in the *lsr* transport genes. The alternative possibility is that AI-2 signals to LsrR from outside the cell. Conceivably, an AI-2 sensor exists that detects external AI-2 and initiates a signalling cascade that culminates in LsrR-mediated derepression of the transcription of the *lsr* operon. We favour an external mechanism for AI-2 signal relay to LsrR because it is more consistent with our results. Specifically, regulation of transcription of the *lsr* operon can occur in the absence of a detectable level of internalization of AI-2. In Fig. 7, we have included a putative AI-2 sensor, and we are currently performing experiments to determine the method of AI-2 signal relay to LsrR. Because LsrR contains a predicted helix–turn–helix DNA-binding domain and because, in bacteria, genes specifying regulatory proteins are frequently located adjacent to the promoter to which they bind, we hypothesize that LsrR represses transcription of the *lsr* operon by binding to the *lsr* promoter. We are testing this now using DNA gel shift analyses.

In an attempt to determine the fate of the AI-2 molecule, we tested whether we could detect internalized AI-2 by lysing various *S. typhimurium luxS* null mutants after incubation with *in vitro*-prepared AI-2. We tested the wild-type *lsrR* and Δ*lsrR S. typhimurium* strains. The latter strain was examined because it imports AI-2 more rapidly than the wild type. Therefore, we reasoned that, if AI-2 is

Table 1. *Salmonella typhimurium* strains and plasmids.

Strain	Relevant genotype	Source
S. typhimurium strains		
14028	Wild type	ATCC
SS007	*luxS*::T-POP	Schauder *et al.* (2001)
MET89	*pqiA*::T-POP	This study
JS14	*lsrA*::MudJ	This study
JS15	*lsrA*::MudJ *luxS*::T-POP	This study
MET442	JS15 with pUC18	This study
MET444	JS15 with pAB15	This study
MET235	*lsrC*::MudJ	This study
MET236	*lsrC*::MudJ *luxS*::T-POP	This study
MET281	MET236 with pUC18	This study
MET276	MET236 with pAB15	This study
MET259	*lsrB*::MudJ	This study
MET260	*lsrB*::MudJ *luxS*::T-POP	This study
MET285	MET260 with pUC18	This study
MET280	MET260 with pAB15	This study
MET233	*lsrF*::MudJ	This study
MET234	*lsrF*::MudJ *luxS*::T-POP	This study
MET231	*lsrF*::MudJ	This study
MET239	*lsrF*::MudJ	This study
MET240	*lsrF*::MudJ *luxS*::T-POP	This study
MET283	MET240 with pUC18	This study
MET278	MET240 with pAB15	This study
MET237	*lsrE*::MudJ	This study
MET238	*lsrE*::MudJ *luxS*::T-POP	This study
MET282	MET238 with pUC18	This study
MET277	MET238 with pAB15	This study
MET328	*rpsL* null	This study
MET341	*rpsL* null Δ*lsrR*	This study
MET342	*rpsL* null Δ*lsrR luxS*::T-POP	This study
MET371	*rpsL* null *lsrC*::MudJ Δ*lsrR luxS*::T-POP	This study
MET370	*rpsL* null *lsrC*::MudJ Δ*lsrR*	This study
MET450	MET236 with pMET1035	This study
MET449	MET235 with pMET1035	This study
MET492	MET371 with pMET1035	This study
MET453	MET370 with pMET1035	This study
MET269	*luxS*::pKAS32	This study
MET270	*lsrC*::MudJ *luxS*::pKAS32	This study
MET301	*lsrC*::MudJ *luxS*::pKAS32 *lsrR*A21T	This study
MET294	*lsrC*::MudJ *luxS*::pKAS32 *lsrR*A120T	This study
MET293	*lsrC*::MudJ *luxS*::pKAS32 *lsrR*G208R	This study
MET299	*lsrC*::MudJ *luxS*::pKAS32 Δ(P_{lsr}-5′ *lsrR*)	This study
MET305	*lsrC*::MudJ *luxS*::pKAS32 *lsrR*L145Q	This study
MET303	*lsrC*::MudJ *luxS*::pKAS32 *lsrR*L134P	This study
MET309	*lsrC*::MudJ *luxS*::pKAS32 *lsrR*L39P	This study
MET312	*lsrC*::MudJ *luxS*::pKAS32 *lsrR*Y25H	This study
MET378	*lsrC*::MudJ *luxS*::pKAS32 *lsrR*::T-POP	This study
JS128	*metE551 metA22 ilv452 trpB2 hisC527(am) galE496 xyl-404 rpsL120 flaA66 hsdL6 hsdA29 zjg8103::pir*	J. Slauch
Plasmids		
pUC18	Amp^R	Sambrook *et al.* (1989)
pAB15	pUC18 (Amp^R) containing *luxS*	This study
pJAF329	pALTER-1 Tet^S, Amp^R	This study
pMET1035	pJAF329 (Amp^R) with *lsrR*	This study
pKAS32	Amp^R, *rpsL* $^+$ suicide vector	Skorupski and Taylor (1996)
pNK2880	Amp^R, P_{tac}-*tnpA*	Bender and Kleckner (1992)

altered immediately after entry into the cells, in the Δ*lsrR* strain, unmodified AI-2 might persist long enough for it to be detected. AI-2 activity could not be detected in lysates of either strain after incubations with AI-2 from 1 min to 1 h. As a control, we performed the lysis experiment with wild-type *S. typhimurium*, which produces AI-2 in the cytoplasm. In this case, we detected significant AI-2 activity in the cell lysates (not shown). Our interpretation of these results is that, upon Lsr-mediated entry into *S. typhimurium*, AI-2 is modified and, therefore, is no longer recognized by the *V. harveyi* AI-2 reporter strain.

We do not understand the benefit that *S. typhimurium* derives from synthesizing and releasing AI-2 only to internalize it at later times. One possibility is that AI-2 is initially used as a signal then, at subsequent times, when AI-2 is no longer required for signal transduction, *S. typhimurium* internalizes it for use as a carbon source. To test this idea, we attempted to grow wild-type *S. typhimurium* on *in vitro*-prepared AI-2 as a sole carbon source. We found that *S. typhimurium* could not grow on AI-2, suggesting that *S. typhimurium* does not use AI-2 as a carbon source (not shown). Therefore, growth on AI-2 cannot account for its disappearance from the medium. A second possibility is that the removal of AI-2 activity is required for its modification to another signal. For example, the LsrE epimerase, or another enzyme, could convert AI-2 into a molecule that has an internal signalling role. Finally, AI-2 could be internalized for degradation. In this case, elimination of AI-2 could serve to terminate the signalling process. Signal production coupled to subsequent elimination could allow *S. typhimurium* the flexibility to transition between several modes of existence. For example, as AI-2 is associated with the presence of many species of bacteria, high levels of AI-2 could be indicative of the transition from outside a host to the intestinal environment, whereas low levels of AI-2 could signal *S. typhimurium* that it has exited the intestinal tract for the next location in the infection (i.e. the macrophage) where *S. typhimurium* is not in association with high numbers of commensal bacteria. Many possibilities exist, the point being that the accumulation and disappearance of AI-2 could each initiate a different series of behavioural changes in *S. typhimurium*.

The genetic screen that led to the identification of the *lsr* operon was designed to reveal the entire collection of genes regulated by AI-2. However, this experiment only yielded the *lsr* operon and *metE*. For reasons discussed above, we do not believe that *metE* is a genuine target of AI-2 regulation. It is possible that the sole target of AI-2 regulation in *S. typhimurium* is the *lsr* operon. Alternatively, we favour the idea that some bias in our screen exists that only allowed us to identify the *lsr* operon. It is possible that *lsr* is the only exclusive target of AI-2. Other AI-2 targets could exist, but their regulation may require AI-2 to act in conjunction with another signal(s). Under the conditions in which we performed our experiments, it is possible that we were not supplying this additional hypothetical signal. Alternatively, two *S. typhimurium* quorum-sensing signals could exist, either of which is sufficient to regulate other target genes. A precedent for redundant signalling factors involving AI-2 exists. In the *V. harveyi* quorum-sensing circuit in which AI-2 was originally discovered, two autoinducers (AI-1 and AI-2) operate in parallel to regulate the expression of bioluminescence. Inactivation of genes involved in either signalling system alone does not abolish the density-dependent expression of luminescence in *V. harveyi* (Bassler *et al*., 1993; 1994a,b). Moreover, nearly all LuxI/LuxR quorum-sensing systems have now been shown to be controlled by multiple autoinducers and, in many cases, intricate hierarchies of regulation exist in these systems (de Kievit and Iglewski, 2000; Miller and Bassler, 2001). The architecture of the *S. typhimurium* quorum-sensing circuit could resemble that of *V. harveyi* or other quorum-sensing bacteria, in that a complex and/or a redundant signalling circuit could govern the process. Finally, it is possible that additional targets of AI-2 are only regulated when *S. typhimurium* is grown in mixed-species consortium. We have proposed that AI-2 is a signal used by many species of bacteria for interspecies cell–cell communication. We assume that interspecies communication is a complex phenomenon that involves multiple sensory inputs. Therefore, AI-2 plus additional signals, cell–cell contact and/or some other stimulus provided by other species of bacteria could be required for *S. typhimurium* to react. We have initiated new genetic and biochemical experiments to identify additional regulators and targets of the Lsr–AI-2 circuit.

Experimental procedures

Bacterial strains and media

The bacterial strains and plasmids used in this study are listed in Table 1. *S. typhimurium* 14028 is ATCC strain 14028, organism: *Salmonella choleraesuis*. All strains were grown in Luria broth (LB), which contained 10 g l^{-1} Bacto tryptone (Difco), 5 gl^{-1} yeast extract (Difco) and 10 gl^{-1} NaCl (Sambrook *et al*., 1989). When necessary, media were supplemented by antibiotics at the following concentrations (mg l^{-1}): ampicillin (Amp), 100; kanamycin (Kan), 100; tetracycline (Tet), 10 or 15; and streptomycin (Sm), 1000. 5-Bromo-4-Chloro-3-indoylyl-β-D-galactoside (Xgal) was added to LB agar plates to a final concentration of 40 mg l^{-1}. MacConkey–lactose and M63–lactose minimal plates were prepared as described previously (Miller, 1992). AB medium was prepared as described previously (Greenberg *et al*., 1979). All *E. coli* and *S. typhimurium* strains were grown at 37°C with aeration.

Genetic and molecular techniques

DNA ligation reactions were transformed by electroporation into either *E. coli* strain JM109 (*supE* Δ(*lac-proAB*) *hsdR17 recA1* F^- *traD36 proAB*$^+$ *lacI*^{q} *lacZ*Δ*M15*) or MC4100 λ*pir* (F^- *araD139* Δ(*argF-lac*)*U169 rpsL150 relA1 thi fib5301 deoC1 ptsF25 rbsR* λ*pir*). Electroporation of plasmids into bacterial cells was carried out as described previously (Sambrook *et al*., 1989). Plasmids were maintained in *E. coli* strains and passaged through the restriction-deficient strain *S. typhimurium* JS128 before introduction into other *S. typhimurium* strains. P22 transductions were performed using P22HT *int* as described previously (Davis *et al*., 1980). Sequencing reactions were performed by the Princeton University SynSeq facility. PCRs were performed using *Taq* DNA polymerase (Boeringer Mannheim Biochemicals) or *Ex-Taq* DNA polymerase (Takara Biochemicals). Restriction endonucleases and T4 DNA ligase were purchased from New England Biolabs. All enzymes were used as recommended by the suppliers. Primers used in PCRs and sequencing reactions are listed in Table 2.

Identification of AI-2-regulated genes in S. typhimurium

The following strategy allowed us to generate 11 000 isogenic wild-type *luxS* and *luxS* null strains harbouring random MudJ (*lacZ*) reporter insertions in the *S. typhimurium* chromosome. Wild-type *S. typhimurium* 14028 was mutagenized with the transposon MudJ by a P22 delivery system, as described previously (Hughes and Roth, 1988). Eleven thousand insertion mutants were ordered onto grids on LB plates that contained Kan to select for MudJ and 5 mM EGTA to inhibit phage adsorption. In order to create the isogenic wild-type *luxS* and *luxS* null pairs of mutants for each MudJ insertion, the mutants were replica plated onto two different Petri plates. One plate contained a P22 lysate that had been propagated on the *luxS* null strain SS007 (14028 *luxS*::T-POP) (Schauder *et al*., 2001). The second plate contained a P22 lysate that was harvested from wild-type *luxS* strain MET89 (14028 *pqiA*::T-POP). The *pqiA* gene is expected to have no effect on AI-2 signalling, and thus serves as a control for the P22 transduction procedure (Koh and Roe, 1995). Both lysates were used at a concentration of 2%. In addition, all Petri plates contained Tet, Kan, Xgal and 5 mM $CaCl_2$. Tet was included in the plates to select for T-POP transductants. Kan was added to ensure that all colonies retained a MudJ insertion. The addition of $CaCl_2$ allowed the cells to be infected with P22 on the Petri plates. Xgal was used as a visual monitor for LacZ activity from the MudJ reporter. The insertion mutants were compared to identify those with differential β-galactosidase activity in the wild-type *luxS* and *luxS* null backgrounds. Colonies containing potential *luxS*-regulated fusions were purified and tested in quantitative β-galactosidase assays. Fusions that showed differential β-galactosidase activity were back-crossed by P22 transduction into wild-type strain 14028 and *luxS* null strain SS007, and subsequently re-examined for differential LacZ activity using β-galactosidase assays. In strains that maintained differential β-galactosidase activity, the chromosome–MudJ transposon fusion junction was amplified by a two-step PCR method. The fusion junction was sequenced in order to identify the regulated gene, as described previously (Surette *et al*., 1999).

Isolation of suppressors of lsr operon expression

Strain MET270 (*luxS*::pKAS32, *lsrC*::MudJ) is Lac^- and is therefore unable to grow on M63–lactose minimal plates. To identify mutations in regulators of the *lsr* operon, a genetic selection was performed for spontaneous mutations in MET270 that resulted in a Lac^+ phenotype, indicating that the *lsrC*::MudJ reporter was transcribed in the absence of a functional *luxS* gene. Mutants were selected by plating aliquots (0.1 ml) of overnight cultures of MET270 onto M63–lactose plates containing Amp and Kan and incubating at 37°C. Amp and Kan were included on the plates to ensure that colonies retained their *luxS*::pKAS32 and *lsrC*::MudJ markers respectively.

In an analogous strategy, transposon T-POP mutagenesis was used to identify mutants in which the *lsr* operon was expressed in a *luxS* null background. A library of 12 000 random T-POP insertions was generated in an *S. typhimurium* strain containing the *lsrC*::MudJ fusion and the Tn*10* transposase-expressing plasmid pNK2880, as described previously (Rappleye and Roth, 1997). A P22 lysate was generated on this library of T-POP insertion mutants to transduce the T-POP insertions into MET270. Transductants were plated on M63–lactose minimal plates containing Amp, Kan and Tet. Amp and Kan were included to ensure retention of the *luxS*::pKAS32 and *lsrC*::MudJ markers respectively. Tet was included to select for T-POP transductants. To verify

Table 2. PCR and sequencing primers.

Primer name	Oligonucleotide
St-11	ATAAGAATGCGGCCGCAGAGGCGTTAAATGACTGCAACGC
St-12	GCGGAGCTCTATCGCTCATTGTCATAACCTGGC
St-13	GCGGAGCTCACTATATCAATGCACTGGTTACCG
St-14	GCGGAATTCAACAGACTACGTTTCCAGTTGCGG
St-15	GCGGAATTCTGAAAAAGAAATTGTTCAAAACGG
Pop1	CCAAATGATGTTATTCCGCG
luxS01	CGGGGATCCTTACCGTAATCTGTTACGCG
luxS04	GGGGATCCGAAAAGCAAGCACCGATCATC
St-18	GCGAAGCTTAGCCAGGTTATGACAATGAGCG
St-16	GCGGGATCCTAATTTGAATTATTTTCCCTGCGG
St-9	AATAAGTATGCGGCCGCCATTCCGAACAAAGAAGTGATG
St-10	GGGGAATTCCGCTGCTCGTCCGGCGTGCCAATC

that the suppressor mutations were associated with the T-POP insertions, linkage of the suppressor mutations to the T-POP insertions was determined by transduction. The T-POP marker was transduced into MET270 and plated on MacConkey–lactose plates containing Amp, Kan and Tet, to select for the *luxS*::pKAS32, *lsrC*::MudJ and T-POP markers respectively. A Lac^+ phenotype was taken as evidence that the suppressor mutation was associated with the T-POP insertion.

To determine whether the suppressor mutations were linked to the *lsr* operon, the *lsrC*::MudJ marker from each mutant was transduced into strain MET269 (*luxS*::pKAS32) and plated on MacConkey–lactose plates containing Kan and Amp. Amp was included in these plates to maintain the *luxS*::pKAS32 construct, and Kan was included to select for MudJ. Co-transduction of the suppressor mutation with the *lsrC*::MudJ fusion was assessed by scoring the presence of Lac^+ colonies, which indicated that the suppressor mutation was linked to the *lsr* operon. To determine whether the T-POP transposon had inserted in *lsrR*, the *lsrR* gene was PCR amplified from the suppressor strains using primers St-11 and St-14. Suppressors generating no PCR product from this reaction were inferred to have a T-POP insertion in the *lsrR* gene. The T-POP–*lsrR* fusion junction was PCR amplified from these strains using T-POP primer Pop1 and *lsrR* primers St-14 and St-15. The products were sequenced to define the positions of the T-POP insertions in *lsrR*. The *lsrR* gene of the spontaneous Lac^+ suppressors was PCR amplified and sequenced to identify the mutations.

Plasmid constructions

To construct plasmid pAB15, the *S. typhimurium luxS* gene and flanking sequences were amplified from the chromosome of *S. typhimurium* 14028 by PCR using primers luxS01 and luxS04. These primers introduce *Bam*HI restriction sites. The resulting PCR product was digested with *Bam*HI and ligated into *Bam*HI-digested pUC18. To construct plasmid pMET1035, the *lsrR* gene was amplified by PCR from the chromosome of *S. typhimurium* 14028 using primers St-18 and St-16, which introduce *Hin*dIII and *Bam*HI restriction sites respectively. The parent vector for this cloning procedure, pJAF329, was created by site-directed mutagenesis of plasmid pALTER-1 (Promega) to create an Amp^R, Tet^S version of the vector. The *lsrR* PCR product was digested with *Hin*dIII and *Bam*HI and ligated into similarly digested pJAF329. Transcription of the *lsrR* gene in this construct is driven from a constitutive promoter in pJAF329.

Construction of the luxS*::pKAS32 null strain*

A 200 bp DNA fragment internal to *luxS* was amplified by PCR from the chromosome of *S. typhimurium* 14028 using primers St-9 and St-10 (Table 2). The PCR product was digested with *Not*I and *Eco*RI and ligated into the suicide plasmid pKAS32 (Skorupski and Taylor, 1996). The resulting *luxS* suicide plasmid was transformed into *S. typhimurium* 14028. Selection on LB plates containing Amp induced the plasmid to integrate into the chromosome at the *luxS* locus. Inactivation of *luxS* was verified by demonstrating that this strain produced no AI-2 in our AI-2 bioassay (Bassler *et al*., 1997). This strain is called MET269.

Construction of an in frame lsrR *deletion*

The DNA flanking the 5′ region of *lsrR* was PCR amplified from *S. typhimurium* 14028 using primers St-11 and St-12. Similarly, the 3′ flanking region was amplified with primers St-13 and St-14. The 5′ sequence was digested with *Not*I and *Sac*I, and the 3′ sequence with *Sac*I and *Eco*RI. The suicide plasmid pKAS32 was digested with *Not*I and *Eco*RI. A three-part ligation was performed in order to clone the *lsrR* flanking regions into pKAS32. This *lsrR* deletion construct was transformed into *E. coli* strain MC4100 λ*pir* by electroporation. The *lsrR* deletion was subsequently transferred to the chromosome of *S. typhimurium* strain MET328 by a two-step suicide delivery system outlined by Skorupski and Taylor (1996).

AI-2 activity assay

Relative levels of AI-2 in cell-free culture fluids of *S. typhimurium* strains of interest were measured using a *V. harveyi* bioluminescence assay, as described previously (Bassler *et al*., 1993; 1997). Cell-free culture fluids were prepared as described (Surette and Bassler, 1998; Surette *et al*., 1999). These preparations were assayed for AI-2 activity at 10% (v/v), using *V. harveyi* reporter strain BB170 (*luxN*::Tn*5*), as described previously (Bassler *et al*., 1993). AI-2 activity is reported as fold induction over background bioluminescence. To examine the role of Lsr in the uptake of AI-2, cultures of *S. typhimurium* were grown overnight, diluted 1:100 into fresh LB and grown for a total of 4 h. During the final hour of growth, the cells were incubated with 70 μM *in vitro*-prepared AI-2 for specified times (Schauder *et al*., 2001). Subsequently, the cultures were chilled on ice, and cell-free culture fluids were prepared. The AI-2 remaining in these preparations was measured in the *V. harveyi* bioluminescence assay. Assays were performed in duplicate.

β-Galactosidase assays

Cultures of *S. typhimurium* that had been grown overnight were diluted 1:100 into fresh LB and grown for 4 h. For strains containing plasmids, cultures were diluted in LB supplemented with Amp. When necessary, AI-2 was synthesized *in vitro* and added to the growth media at the time of inoculation (Schauder *et al*., 2001). Cells from 1 ml of these cultures were harvested and resuspended in 1 ml of Z buffer. β-Galactosidase assays were performed using a microtitre plate assay, as described previously (Slauch and Silhavy, 1991). β-Galactosidase units were calculated as $(OD_{420}\,min^{-1} \times 10^{-4})/[OD_{600} * volume\ (ml)]$. All assays were performed in triplicate.

Acknowledgements

The National Science Foundation grants MCB-0083160 and MCB-0094447 and The Office of Naval Research grant number N00014-99-0767 supported this work. We thank

Dr Stephan Schauder for supplying *in vitro*-prepared AI-2, and Dr Michael Surette for giving us plasmid pAB15. We are especially grateful to Dr James Slauch for his insights about how to perform the phage P22 transductions directly on Petri plates. We also thank him for the generous gift of strain JS128. We thank Dr Thomas Silhavy and Dr Mark Rose for many helpful discussions.

References

Altschul, S.F., Gish, W., Miller, W., Myers, E.W., and Lipman, D.J. (1990) Basic local alignment search tool. *J Mol Biol* **215**: 403–410.

Bassler, B.L. (1999) How bacteria talk to each other: regulation of gene expression by quorum sensing. *Curr Opin Microbiol* **2**: 582–587.

Bassler, B.L., Wright, M., Showalter, R.E., and Silverman, M.R. (1993) Intercellular signalling in *Vibrio harveyi*: sequence and function of genes regulating expression of luminescence. *Mol Microbiol* **9**: 773–786.

Bassler, B.L., Wright, M., and Silverman, M.R. (1994a) Multiple signalling systems controlling expression of luminescence in *Vibrio harveyi*: sequence and function of genes encoding a second sensory pathway. *Mol Microbiol* **13**: 273–286.

Bassler, B.L., Wright, M., and Silverman, M.R. (1994b) Sequence and function of LuxO, a negative regulator of luminescence in *Vibrio harveyi*. *Mol Microbiol* **12**: 403–412.

Bassler, B.L., Greenberg, E.P., and Stevens, A.M. (1997) Cross-species induction of luminescence in the quorum-sensing bacterium *Vibrio harveyi*. *J Bacteriol* **179**: 4043–4045.

Bell, A.W., Buckel, S.D., Groarke, J.M., Hope, J.N., Kingsley, D.H., and Hermodson, M.A. (1986) The nucleotide sequences of the *rbsD*, *rbsA*, and *rbsC* genes of *Escherichia coli* K12. *J Biol Chem* **261**: 7652–7658.

Bender, J., and Kleckner, N. (1992) IS*10* transposase mutations that specifically alter target site recognition. *EMBO J* **11**: 741–750.

Blattner, F.R., Plunkett, G., III, Bloch, C.A., Perna, N.T., Burland, V., Riley, M., *et al.* (1997) The complete genome sequence of *Escherichia coli* K-12. *Science* **277**: 1453–1474.

Boos, W., and Lucht, J.M. (1996) Periplasmic binding protein-dependent ABC transporters. In *Escherichia coli and Salmonella typhimurium: Cellular and Molecular Biology*, **Vol. 1**. Neidhardt, F. (ed.). Washington, DC: American Society for Microbiology Press, pp. 1175–1209.

Buckel, S.D., Bell, A.W., Rao, J.K., and Hermodson, M.A. (1986) An analysis of the structure of the product of the rbsA gene of *Escherichia coli* K12. *J Biol Chem* **261**: 7659–7662.

Caetano-Anolles, G. (1993) Amplifying DNA with arbitrary oligonucleotide primers. *PCR Methods Appl* **3**: 85–94.

Cao, J.G., and Meighen, E.A. (1989) Purification and structural identification of an autoinducer for the luminescence system of *Vibrio harveyi*. *J Biol Chem* **264**: 21670–21676.

Davis, R.W., Botstein, D., and Roth, J.R. (1980) *Advanced Bacterial Genetics.* Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press..

Day, W.A., Jr, and Maurelli, A.T. (2001) *Shigella flexneri* LuxS quorum-sensing system modulates *virB* expression but is not essential for virulence. *Infect Immun* **69**: 15–23.

Engebrecht, J., and Silverman, M. (1984) Identification of genes and gene products necessary for bacterial bioluminescence. *Proc Natl Acad Sci USA* **81**: 4154–4158.

Engebrecht, J., and Silverman, M. (1987) Nucleotide sequence of the regulatory locus controlling expression of bacterial genes for bioluminescence. *Nucleic Acids Res* **15**: 10455–10467.

Engebrecht, J., Nealson, K., and Silverman, M. (1983) Bacterial bioluminescence: isolation and genetic analysis of functions from *Vibrio fischeri*. *Cell* **32**: 773–781.

Freeman, J.A., and Bassler, B.L. (1999a) A genetic analysis of the function of LuxO, a two-component response regulator involved in quorum sensing in *Vibrio harveyi*. *Mol Microbiol* **31**: 665–677.

Freeman, J.A., and Bassler, B.L. (1999b) Sequence and function of LuxU: a two-component phosphorelay protein that regulates quorum sensing in *Vibrio harveyi*. *J Bacteriol* **181**: 899–906.

Freeman, J.A., Lilley, B.N., and Bassler, B.L. (2000) A genetic analysis of the functions of LuxN: a two-component hybrid sensor kinase that regulates quorum sensing in *Vibrio harveyi*. *Mol Microbiol* **35**: 139–149.

Fuqua, C., Winans, S.C., and Greenberg, E.P. (1996) Census and consensus in bacterial ecosystems: the LuxR–LuxI family of quorum-sensing transcriptional regulators. *Annu Rev Microbiol* **50**: 727–751.

Greenberg, E.P., Hastings, J.W., and Ulitzer, S. (1979) Induction of luciferase synthesis in *Beneckea harveyi* by other marine bacteria. *Arch Microbiol* **120**: 87–91.

Holland, I.B., and Blight, M.A. (1999) ABC-ATPases, adaptable energy generators fuelling transmembrane movement of a variety of molecules in organisms from bacteria to humans. *J Mol Biol* **293**: 381–399.

Hope, J.N., Bell, A.W., Hermodson, M.A., and Groarke, J.M. (1986) Ribokinase from *Escherichia coli* K12. Nucleotide sequence and overexpression of the *rbsK* gene and purification of ribokinase. *J Biol Chem* **261**: 7663–7668.

Hughes, K.T., and Roth, J.R. (1988) Transitory *cis* complementation: a method for providing transposition functions to defective transposons. *Genetics* **119**: 9–12.

Iida, A., Harayama, S., Iino, T., and Hazelbauer, G.L. (1984) Molecular cloning and characterization of genes required for ribose transport and utilization in *Escherichia coli* K-12. *J Bacteriol* **158**: 674–682.

de Kievit, T.R., and Iglewski, B.H. (2000) Bacterial quorum sensing in pathogenic relationships. *Infect Immun* **68**: 4839–4849.

Kim, S.Y., Lee, S.E., Kim, Y.R., Kim, J.H., Ryu, P.Y., Chung, S.S., and Rhee, J.H. (2000) Virulence regulatory role of *luxS* quorum sensing system in *Vibrio vulnificus*. In *American Society for Microbiology General Meeting*. Abstract B-248, pp. 97–98

Kleerebezem, M., Quadri, L.E., Kuipers, O.P., and de Vos, W.M. (1997) Quorum sensing by peptide pheromones and two-component signal-transduction systems in Gram-positive bacteria. *Mol Microbiol* **24**: 895–904.

Koh, Y.S., and Roe, J.H. (1995) Isolation of a novel paraquat-inducible (*pqi*) gene regulated by the *soxRS* locus in *Escherichia coli*. *J Bacteriol* **177**: 2673–2678.

Lazazzera, B.A., and Grossman, A.D. (1998) The ins and outs of peptide signaling. *Trends Microbiol* **6**: 288–294.

Lilley, B.N., and Bassler, B.L. (2000) Regulation of quorum sensing in *Vibrio harveyi* by LuxO and sigma-54. *Mol Microbiol* **36**: 940–954.

Lyon, W.R., Madden, J.C., Levin, J.C., Stein, J.L., and Caparon, M.G. (2001) Mutation of *luxS* affects growth and virulence factor expression in *Streptococcus pyogenes*. *Mol Microbiol* **42**: 145–147.

Mauzy, C.A., and Hermodson, M.A. (1992) Structural and functional analyses of the repressor, RbsR, of the ribose operon of *Escherichia coli*. *Protein Sci* **1**: 831–842.

Miller, J.H. (1992) *A Short Course in Bacterial Genetics.* Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press.

Miller, M.B., and Bassler, B.L. (2001) Quorum sensing in bacteria. *Annu Rev Microbiol* **55**: 165–199.

Nikaido, H., and Hall, J.A. (1998) Overview of bacterial ABC transporters. *Methods Enzymol* **292**: 3–20.

Park, Y., Cho, Y.J., Ahn, T., and Park, C. (1999) Molecular interactions in ribose transport: the binding protein module symmetrically associates with the homodimeric membrane transporter. *EMBO J* **18**: 4149–4156.

Rappleye, C.A., and Roth, J.R. (1997) A Tn*10* derivative (T-POP) for isolation of insertions with conditional (tetracycline-dependent) phenotypes. *J Bacteriol* **179**: 5827–5834.

Sambrook, J., Fritsch, E.F., and Maniatis, T. (1989) *Molecular Cloning: A Laboratory Manual.* Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press.

Schauder, S., and Bassler, B.L. (2001) The languages of Bacteria. *Genes Dev* **151**: 1468–1480.

Schauder, S., Shokat, K., Surette, M.G., and Bassler, B.L. (2001) The LuxS family of bacterial autoinducers: biosynthesis of a novel quorum sensing signal molecule. *Mol Microbiol* **41**: 463–476.

Skorupski, K., and Taylor, R.K. (1996) Positive selection vectors for allelic exchange. *Gene* **169**: 47–52.

Slauch, J.M., and Silhavy, T.J. (1991) *cis*-acting *ompF* mutations that result in OmpR-dependent constitutive expression. *J Bacteriol* **173**: 4039–4048.

Sperandio, V., Mellies, J.L., Nguyen, W., Shin, S., and Kaper, J.B. (1999) Quorum sensing controls expression of the type III secretion gene transcription and protein secretion in enterohemorrhagic and enteropathogenic *Escherichia coli*. *Proc Natl Acad Sci USA* **96**: 15196–15201.

Sprenger, G.A. (1995) Genetics of pentose-phosphate pathway enzymes of *Escherichia coli* K-12. *Arch Microbiol* **164**: 324–330.

Surette, M.G., and Bassler, B.L. (1998) Quorum sensing in *Escherichia coli* and *Salmonella typhimurium*. *Proc Natl Acad Sci USA* **95**: 7046–7050.

Surette, M.G., and Bassler, B.L. (1999) Regulation of autoinducer production in *Salmonella typhimurium*. *Mol Microbiol* **31**: 585–595.

Surette, M.G., Miller, M.B., and Bassler, B.L. (1999) Quorum sensing in *Escherichia coli*, *Salmonella typhimurium*, and *Vibrio harveyi*: a new family of genes responsible for autoinducer production. *Proc Natl Acad Sci USA* **96**: 1639–1644.

Urbanowski, M.L., and Stauffer, G.V. (1989) Genetic and biochemical analysis of the MetR activator-binding site in the *metE metR* control region of *Salmonella typhimurium*. *J Bacteriol* **171**: 5620–5629.

Wehmeier, U.F., and Lengeler, J.W. (1994) Sequence of the sor-operon for L-sorbose utilization from *Klebsiella pneumoniae* KAY2026. *Biochim Biophys Acta* **1208**: 348–351.

Weissbach, H., and Brot, N. (1991) Regulation of methionine synthesis in *Escherichia coli*. *Mol Microbiol* **5**: 1593–1597.